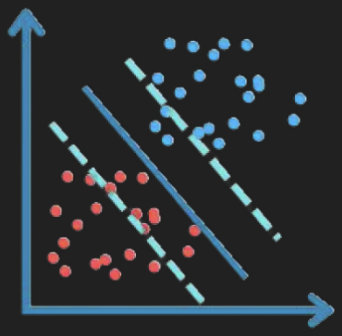
SVMs

Support Vector Machine

SVMs

- Popular in 2000's (late 90s)
- Kernel SVM
- Theoretically they are best
- Less frequently used nowadays
- Challenging Maths



Geometrical Intuition

Idea : Find a line/hyperplane that best separate given classes.

Suppose,

We have data points from both ***+ve*** and ***-ve*** class.



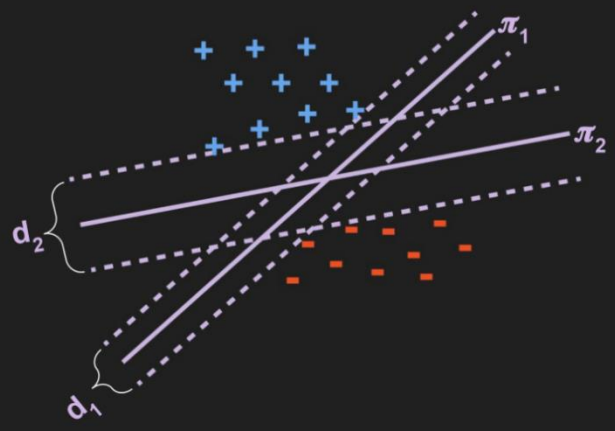
Using a line/hyperplane

Hyperplanes are
decision boundaries
in higher dimensions.

But there can be multiple lines/hyperplanes…

Which line/hyperplane should we choose?

Imagine two hyperplanes π_1 and π_2



$\because$ Margin d_1 <<< Margin d_2

$\therefore$ Pick π_2 s.t. we get max/largest margin

Why do we need a large margin?

Because,

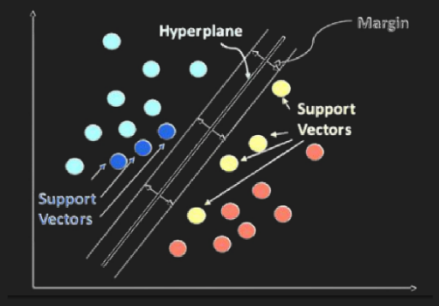
Larger the margin,

Better the separation

between the *+ve* and *-ve* samples.

Such classifiers where,

The margin should be as large as possible
are called **margin-maximizing classifiers.**



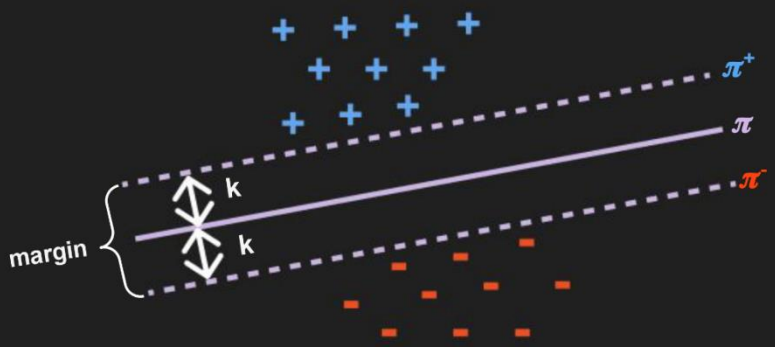
NOTE: Distances from data points are measured perpendicular to any of the hyperplanes.

Assume we have :

- +ve and -ve data points
- π is a margin-maximizing hyperplane
- π^+ is the +ve hyperplane parallel to
- π^+ is touching the closest +ve points to π
- π^- is the -ve hyperplane parallel to
- π^- is touching the closest -ve points to π
- Margin is the dist. between π+ and π-



How do we define the hyperplanes?



Let : $\pi \rightarrow w^Tx + b = 0$

$\pi^+ \rightarrow w^Tx + b = k \rightarrow w^Tx + b - k = 0$

$\pi^- \rightarrow w^Tx + b = -k \rightarrow w^Tx + b + k = 0$

What will be the length of margin?

According to linear algebra, distance from origin :-

For : $\pi^+ : d(0, \pi^+) = \frac{b - k}{||w||}$

For : $\pi^- : d(0, \pi^-) = \frac{b + k}{||w||}$

So,

$$margin\ d(\pi^+, \pi^-) = d(0, \pi^-) - d(0, \pi^+) = \frac{2k}{||w||}$$

What are the parameters of the margin?

- weight **(w)**
- constant **(b)**

Consider these two cases :

Case 1

k = 1

$$d = \frac{2 * 1}{||w||} = \frac{2}{||w||}$$

$$max_{w,b}(\frac{2}{||w||})$$

Case 2

k = 10

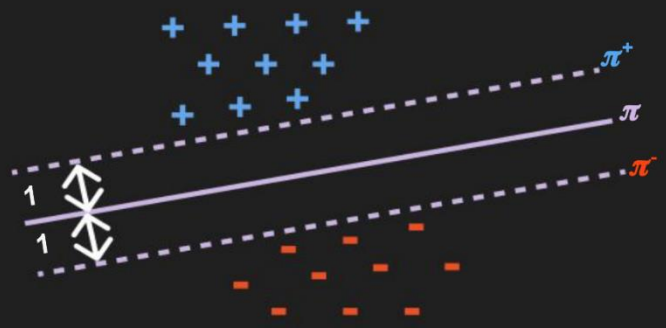
$$d = \frac{2 * 10}{||w||} = \frac{20}{||w||}$$

$$max_{w,b}(\frac{20}{||w||})$$

Changing the value of (k) would only scale the margin but won't affect the position of the hyperplane.

Hence, for mathematical simplicity we take k=1

Therefore, with k=1



$\pi^+ : w^Tx + b - 1 = 0$

$\pi^- : w^Tx + b + 1 = 0$

$d(\pi^+, \pi^-) = (\frac{2}{||w||})$

Now, our goal is to :

- Maximize the margin $(\frac{2}{||w||})$
- With respect to (w) and (b) to obtain best possible separation.

Hard Margin SVM

In a Binary Classification problem,

y_i = +1 for +ve samples

y_i = -1 for -ve samples

Now the **functional margin** i.e. the signed distance between

- the data point x_i and
- the hyperplane π is…

  $= y_i(w^Tx + b)$



Hard Margin SVM

Our goal is to maximize the margin
s.t. the +ve and -ve samples are on different sides of hyperplane π

Therefore, the margin constraints are :

For $y_i = +1 \longrightarrow \pi^+ = y_i (w^T x + b) \geq 1$

For $y_i = -1 \longrightarrow \pi^- = y_i (w^T x + b) \leq -1$

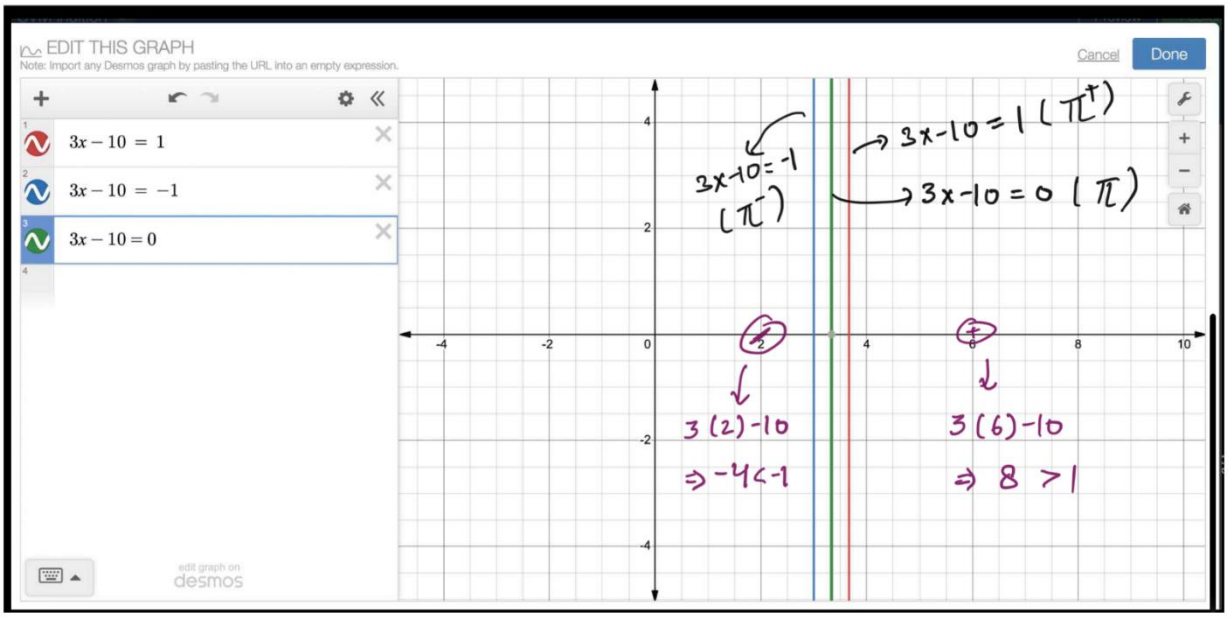
If we club both these equations:

$$argmax_{w,b}\left(\frac{2}{||w||}\right) \; s.t. \; y_i(w^T x_i + b) \geq 1 \quad \forall \; i : 1 \longrightarrow N$$

How does $y_i \; (w^T x + b) \geq 1$ works?

For *+ve* samples	For *-ve* samples
$y_i = +1$ (+ve)	$y_i = -1$ (-ve)
$w^T x_i + b \geq 1$ (+ve)	$w^T x_i + b \leq -1$ (-ve)
+ve * +ve = +ve	-ve * -ve = +ve





Why the $y_i\,(w^Tx + b) \geq 1$ constraint?

Our goal is to:

- completely seperate the two classes
- with a margin as large as possible

With this constraint:

- all *+ve* data points should lie beyond π^+
- all *-ve* data points should lie beyond π^-

This is called **Hard Margin Classification**

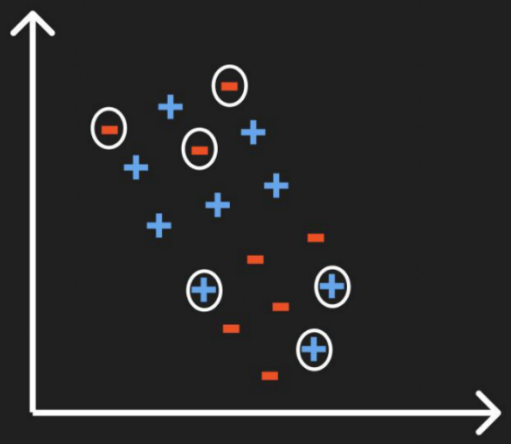


Which means, we strictly impose that:

All instances must be

- off the margin and
- on the correct side
- i.e., Zero Error

When would a linear SVM with hard margin fail?



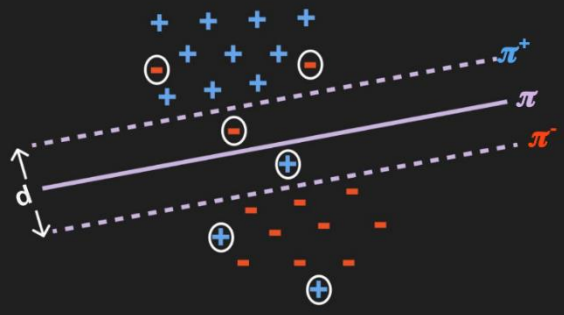
Data is not linearly separable and hence it has some some outliers.

Hard Margin SVM fails

Soft Margin SVM

What if the data isn't perfectly linearly separable?

Some data points lie on the wrong side of the hyperplane π.



Almost linearly separable

Our goal is to :

- Maximize the margin
- Minimize data points with error $\zeta_i > 0$
- Minimize the errors ζ_i 's

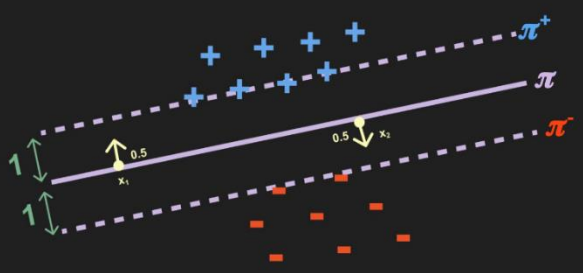
How to account for these data points?

Imagine,

- a +ve data point x_1 at 0.5 unit distance between π & π^+
- a -ve data point x_2 at 0.5 unit distance between π & π^-

$y_1(w^T x_1 + b) = 0.5 = 1 - 0.5 \rightarrow \zeta_1 = 0.5$

$y_2(w^T x_2 + b) = 0.5 = 1 - 0.5 \rightarrow \zeta_2 = 0.5$



In general,

$$y_i(w^T x_i + b) \geq 1 - \zeta_i$$

Note:

- $\zeta_i = 0$ for all correctly placed points
- $\zeta_i > 0$ for all misclassified points

→ $\zeta_i \geq 0$ for all data points

Now, the optimization fn changes to :

$$min_{w,b} \frac{||w||}{2} + \frac{C}{N} \sum_{i=1}^{N} \zeta_i \quad s.t. (w^T x_i + b) y_i \geq 1 - \zeta_i$$

$$\forall\, i : 1 \rightarrow N \; and \; \zeta_i \geq 0$$

This is known as Soft Margin SVM

What's the use of 'C' here?

C is a hyperparameter.
It controls whether to focus on :

- **Maximizing margin**
- **Minimizing errors ζ_i 's**

Case 1

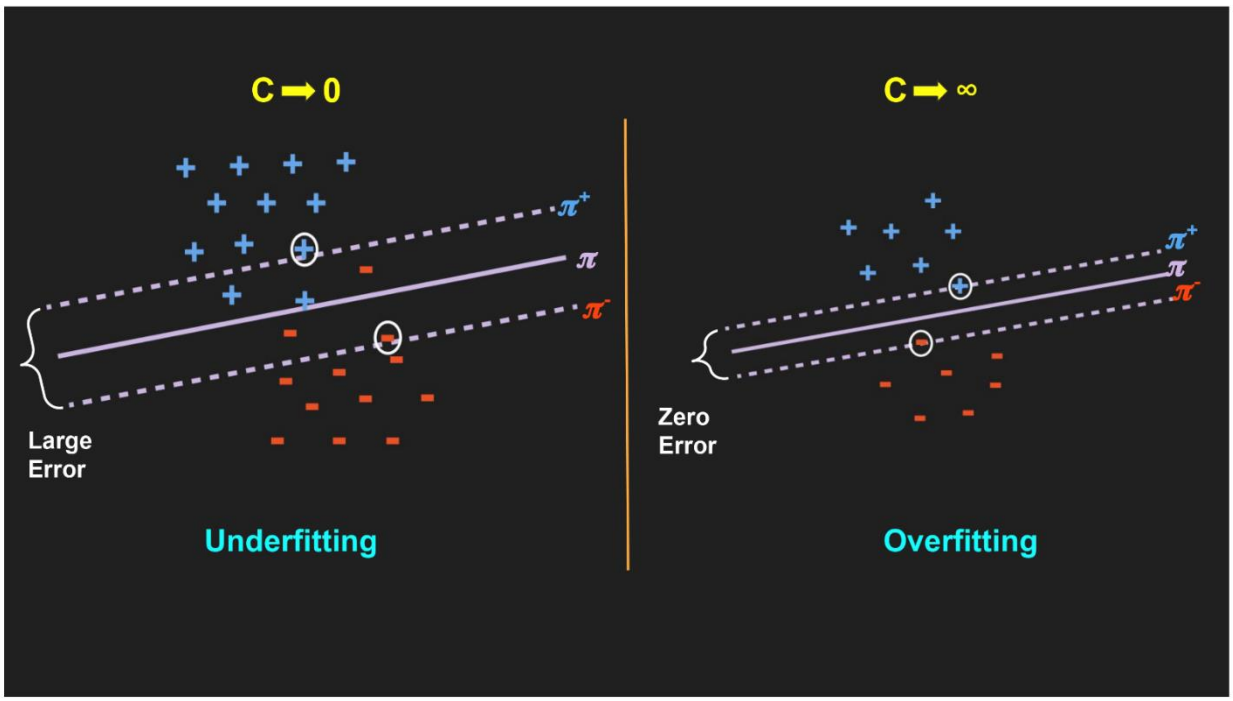
As C ↓

- Model becomes too lenient towards misclassifications.
- More importance is given to maximizing margins.
- The model may Underfit.

Case 2

As C ↑

- Model becomes less lenient towards misclassifications.
- More importance is given to minimizing errors.
- The model may Overfit.



Algebraic Intuition

We saw that **soft margin SVMs** are defined as :

$$min_{w,b} \frac{||w||}{2} + C \frac{1}{N} \sum_{i=1}^{N} \zeta_i$$ → Hinge Loss

The term $min_{w,b} \frac{||w||}{2} = L2$ Regularization and

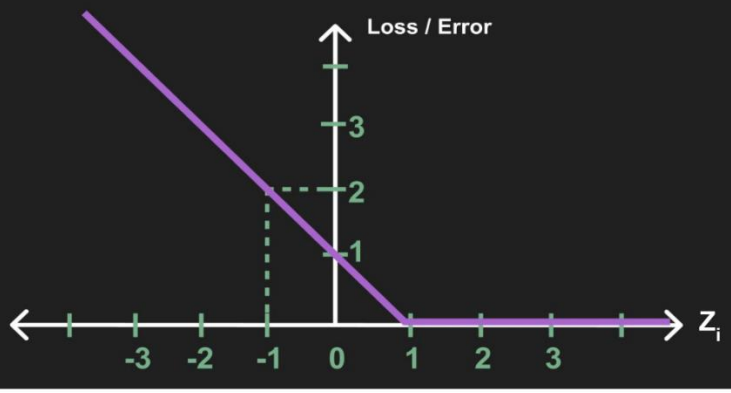
(C) is analogous to regularization hyperparam (λ)

∴ Soft Margin SVM = *½ L2 Reg + C Hinge Loss*

Intuition of Hinge Loss

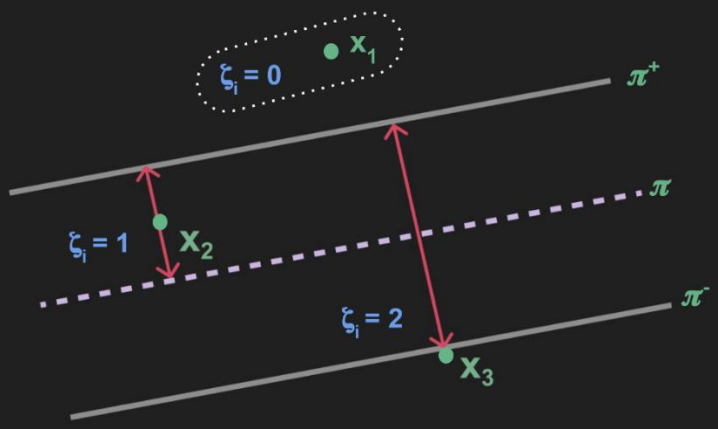
Let's define $(w^T x + b)$ as $f(x_i)$

Assume X-axis to be : $Z_i = y_i(f(x_i)) = y_i(w^T x_i + b)$



Consider three data points x_1 , x_2 , x_3 that belong to +ve class.





Case 1

- Point x_1 , lies on/beyond π^+
- Then $Z_i \geq 1$

  $Y_i (w^T x + b) \geq 1$
- So, $\zeta_i = 0$

  $d (x_1 , \pi^+) = 0$

Case 2

- Point x_2 , lies on π
- Then $Z_i = 0$

  $Y_i (w^T x + b) = 0$
- So, $\zeta_i = 1$

  $d (x_2 , \pi^+) = 1$

Case 3

- Point x_3 , lies on π^-
- Then $Z_i = -1$

  $Y_i (w^T x + b) = -1$
- So, $\zeta_i = 2$

  $d (x_3 , \pi^+) = 2$





What can we conclude from this?



- Error cannot be negative. $\zeta_i \geq 0$
- According to the constraint -

  $y_i (w^T x + b) > - 1 - \zeta$

  $= Z_i > 1 - \zeta$

  $= \zeta_i \geq 1 - \zeta - Z_i$

We can combine these two equations and get Hinge Loss $\zeta_i = max (0 , 1 - Z_i)$

NOTE: As Z_i ↑ ζ_i ↓ till zero

As Z_i ↓ ζ_i ↑

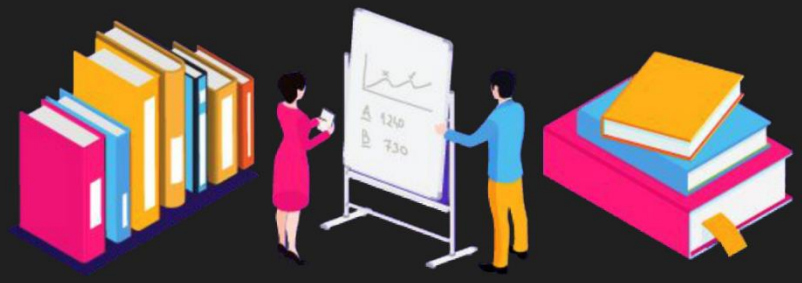
Are SVMs affected by class imbalance?

Only a few data points contribute to the **Hinge loss (ζ_i)** .

These points are called **Support Vectors**.

Hence, SVM is only affected by **imbalance in no. of support vectors** for each class.

NOTE: The balance in no. of support vectors can't be guaranteed.



I feel it will be better to call it a multiplier by which we control how harshly we treat misclassifications, with c=0 meaning no penalty to the model for misclassifications and c= 《very high number》 meaning very heavy penalty to model for any misclassification.